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For Performance Measurement

**ZIMBABWE SCHOOL EXAMINATIONS COUNCIL**

General Certificate of Education Advanced Level

**CHEMISTRY****6031/1**

PAPER 1 Multiple Choice

**TIME: 1 hour**

## Additional materials:

- Data Booklet
- Mathematical tables and/or Electronic calculator
- Multiple Choice answer sheet
- Soft pencil (type B or HB is recommended)

**INSTRUCTIONS TO CANDIDATES****Do not open this booklet until you are told to do so.**

Write your name, Centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**. Choose the one you consider correct and record your choice in soft pencil on the separate answer sheet.

**Read very carefully the instructions on the answer sheet.****INFORMATION FOR CANDIDATES**

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

**This question paper consists of 16 printed pages.**

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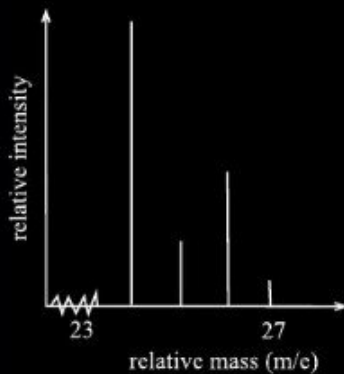
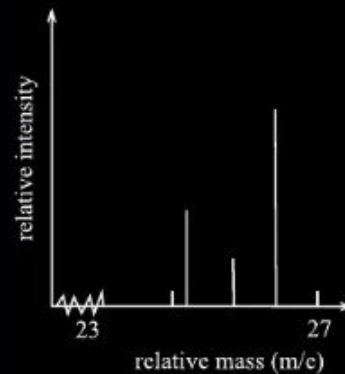
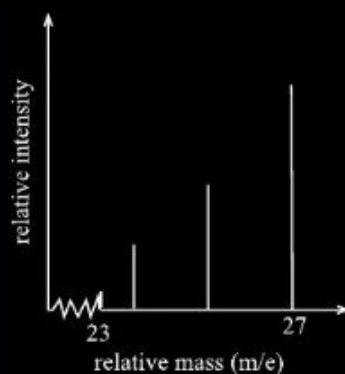
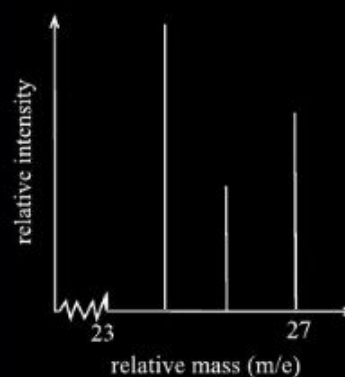


## Section A

For each question there are four possible answers, A, B, C and D. Choose the one you consider to be correct.

- 1 The average mass ratio of the isotopes of the element  ${}^{24.3}\text{X}$  in increasing order is 7 : 1 : 2.

What is the mass spectrum of X?

**A****B****C****D**



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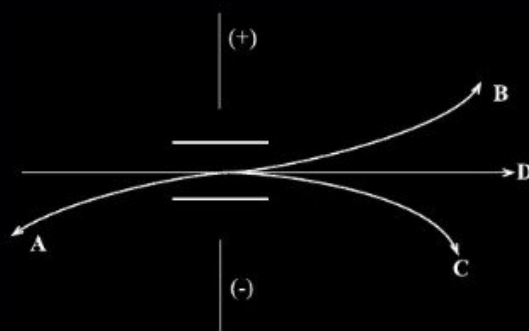


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3

- 2 Which letter, A, B, C or D shows the correct behaviour of proton beams in an electric field?



- 3 Which molecule is pyramidal?

- A  $\text{SO}_2$   
 B  $\text{PCl}_3$   
 C  $\text{BF}_3$   
 D  $\text{AlCl}_3$

- 4 What is the number of moles of a gas occupying a volume of  $0.25 \text{ m}^3$  at  $1.01 \times 10^5 \text{ Pa}$  and  $565^\circ\text{C}$ ?

A  $\frac{1.01 \times 10^5 \times 0.25}{8.31 \times 838}$

B  $\frac{1.01 \times 10^5 \times 0.25}{8.31 \times 565}$

C  $\frac{8.31 \times 838}{1.01 \times 10^5 \times 0.25}$

D  $\frac{8.31 \times 565}{1.01 \times 10^5 \times 0.25}$



5 Which property of Group 7 hydrides increases down the group?

- A acidity
- B boiling point
- C bond energy
- D thermal stability

6 Which cation A, B, C or D is most oxidising?

- A  $\text{Pb}^{2+}$
- B  $\text{Cr}^{3+}$
- C  $\text{Fe}^{3+}$
- D  $\text{Sn}^{2+}$

7 Which acid A, B, C or D is the strongest?

|   | acid            | $K_a$ value          |
|---|-----------------|----------------------|
| A | HF              | $3.5 \times 10^{-4}$ |
| B | $\text{HClO}$   | $29 \times 10^{-8}$  |
| C | $\text{HClO}_2$ | $1.1 \times 10^{-2}$ |
| D | HCN             | $49 \times 10^{-10}$ |

8 The measured initial rates for the reaction,  $\text{X}_{(\text{g})} + \text{Y}_{(\text{g})} \rightarrow \text{Z}_{(\text{g})} + \text{W}_{(\text{g})}$ , for different concentrations of the reactants are shown.

|  |      |      | initial rate / |
|--|------|------|----------------|
|  | 0.10 | 0.10 | 0.002          |
|  | 0.20 | 0.10 | 0.008          |
|  | 0.20 | 0.20 | 0.008          |

What is the rate expression for this reaction?

- A  $k[\text{Y}]^2$
- B  $k[\text{X}][\text{Y}]$
- C  $k[\text{X}]^2[\text{Y}]^2$
- D  $k[\text{X}]^2$

5

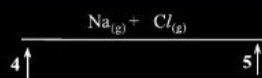
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9 Which of the elements A, B, C or D is most reactive with water?

- A Mg
- B Be
- C Sr
- D Ca



10 The diagram shows a Born-Haber cycle for the formation of  $\text{NaCl}_{(s)}$ .



Which set correctly describes the enthalpy changes 1 and 3?

- |   | 1         | 3           |
|---|-----------|-------------|
| A | lattice   | atomisation |
| B | formation | atomisation |
| C | formation | ionisation  |
| D | lattice   | ionisation  |

11 How many  $\sigma$  and  $\pi$  bonds are in a benzene molecule

- A 6 3
- B 12 3
- C 6 6
- D 12 12

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12 Which reaction is feasible under standard conditions?

- A  $Al_{(s)} + 3Ag_{(aq)}^+ \rightarrow Al_{(aq)}^{3+} + 3Ag_{(s)}$   
 B  $Ni_{(aq)}^{2+} + Pb_{(s)} \rightarrow Ni_{(s)} + Pb_{(aq)}^{2+}$   
 C  $Fe_{(s)} + Mg_{(aq)}^{2+} \rightarrow Fe_{(aq)}^{2+} + Mg_{(s)}$   
 D  $Zn_{(s)} + Ca_{(aq)}^{2+} \rightarrow Zn_{(aq)}^{2+} + Ca_{(s)}$

13 Which molecule or ion, A, B, C or D can act as a ligand?

- A  $H_2O$   
 B  $NO_2^+$   
 C  $NH_4^+$   
 D  $HCN$

14 Ethanoic acid dissociates according to the equation:



What is the effect of adding sodium ethanoate to the equilibrium mixture?

- A equilibrium shifts to the right  
 B more  $CH_3COO^-$  ions are produced  
 C  $H_3O^+$  ion concentration increases  
 D  $H_3O^+$  ion concentration decreases

15 Which atomic size relationship A, B, C or D is correct?

- A  $P < S$   
 B  $P > Se$   
 C  $S < Cl$   
 D  $S < Se$

16 Which statement explains why the first ionisation energy of sulphur is less than that of phosphorus?

- A The electron removed occupies a subshell at a higher energy level.  
 B The electron is at a greater distance from the nucleus.  
 C The electron experiences smaller effective nucleus charge.  
 D The electron experiences greater electron repulsion.

17 Which combination would result in a displacement reaction?



- A  $I_{2(s)} + NaBr_{(aq)}$   
 B  $Cl_{2(g)} + NaI_{(aq)}$   
 C  $Br_{2(l)} + NaCl_{(aq)}$   
 D  $I_{2(s)} + NaCl_{(aq)}$

18 Nitrogen gas is used to manufacture ammonia by the Haber process as shown:



Which statement about the reaction is correct?

- A 28g of  $N_2$  produces 1 mole of  $NH_3$   
 B low pressure favours the production of  $NH_3$   
 C low temperature favours the production of  $NH_3$   
 D high temperature favours the production of  $NH_3$
- 19 Which electrochemical cell notation is correct for the reaction



- A  $MnO_4^{-}(aq) / Mn^{2+}(aq) // Fe_{(s)} / Fe^{2+}(aq)$   
 B  $Fe_{(s)} / Fe^{2+}(aq) // MnO_4^{-}(aq) / Mn^{2+}(aq)$   
 C  $Mn^{2+}(aq), H^{+}(aq) / MnO_4^{-}(aq) // Fe^{2+}(aq) / Fe_{(s)}$   
 D  $Fe_{(s)} / Fe^{2+}(aq) // MnO_4^{-}(aq), H^{+}(aq) / Mn^{2+}(aq) / Pt$

20 Which pair A, B, C or D are isomers?

- A propene, methylpropane  
 B propanone, ethanol  
 C propanal, propanone  
 D propanal, diethylether

21 The structure of alanine is shown

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21 The structure of alanine is shown

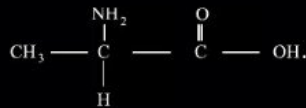
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8



Which is a property of alanine?

- A undergoes acid base reaction
- B dissolves in non polar solvents
- C has a low melting point
- D the molecule is achiral

22 Which molecule exhibits geometrical isomerism?

- A 2,4 - dichloro - 2 - but - 2 - ene
- B 1,4 - dichlorobenzene.
- C 4-methylhex - 2 - ene
- D 1,1-dichloro - but - 1 - ene



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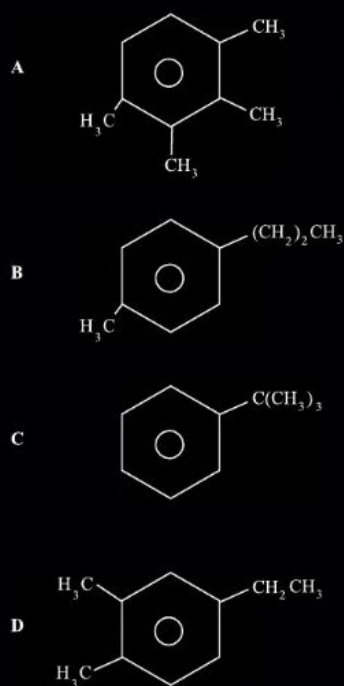
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9

- 23 An organic compound of the molecular formula  $C_{10}H_{14}$  gives two products on aromatic substitution by chlorine.

The organic compound could be



- 24 Propanol reacts with Tollen's reagent to produce

- A silver and propanol.
- B silver oxide and propanoic acid.
- C silver nitrate and propanoic acid.
- D silver and propanoic acid.

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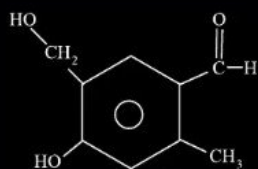


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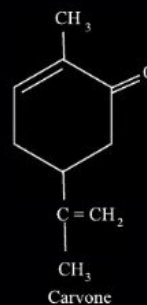


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- 25 The structural formula of vanillin and carvone are shown:



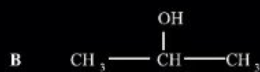
Vanillin



Carvone

Which reagents can be used to distinguish between vanillin and carvone?

- A aqueous bromine
  - B Fehling's solution
  - C alkaline aqueous bromine
  - D 2,4 dinitro-phenylhydrazine
- 26 Which compound, A, B, C or D is easily oxidised?



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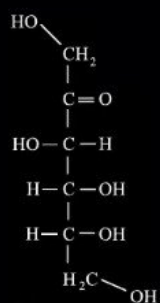
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11

- 27 The diagram shows the structure of fructose.



fructose

How many chiral carbon atoms are in fructose?

- A 1  
B 2  
C 3  
D 6
- 28 Which method of waste disposal is the most recommended?
- A incineration  
B bioremediation  
C reusing  
D land filling

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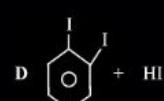
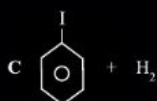
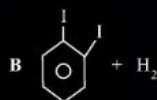
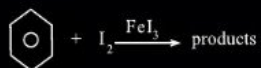


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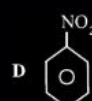
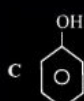
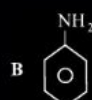
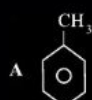
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29 What are the correct products of the reaction shown?



30 Which benzene derivative A, B, C or D is least reactive?



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## Section B



For each of the questions in this section, one or more of the three numbered statements 1 to 3 may be correct.

Decide whether each of the statements is or is not correct. (You may find it helpful to put a tick against the statement(s) which you consider to be correct).

The responses A to D should be selected on the basis of

| A                      | B                        | C                        | D                 |
|------------------------|--------------------------|--------------------------|-------------------|
| 1, 2 and 3 are correct | 1 and 2 only are correct | 2 and 3 only are correct | 1 only is correct |

No other combination of statements is used as a correct response.

31 Which factor(s) affect(s) the standard electrode potential of a cell?

1. pressure
2. concentration
3. temperature

32 Which species has/ have an oxidation number of +3?

1.  $\text{Al}_2\text{O}_3$
2.  $\text{P}_4\text{O}_6$
3.  $\text{Cr}_2\text{O}_7^{2-}$

33 Which organic substance rotates plane polarised light?

1.  $\text{CH}_3(\text{CHOH})_2\text{CH}_3$   
 $\text{CH}_3\text{CHBrCH}_2\text{CH}_3$   
 $\text{CH}_3\text{CHOHCHClCH}_3$

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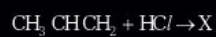
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14

34 The reaction of a hydrocarbon is shown:



What could be X?

1. 2 - chloropropane
2. a mixture of 2 - chloropropane and 1 - chloropropane
3. 1 - chloropropane

35 Which property of period 3 elements decreases with increasing atomic number?

1. atomic radii
2. easy of losing an electron by an atom
3. acidity of the oxides

36 Which species is/are nucleophile(s)

1.  $\text{NH}_4^+$
2.  $\text{NH}_3$
3.  $\text{HSO}_4$

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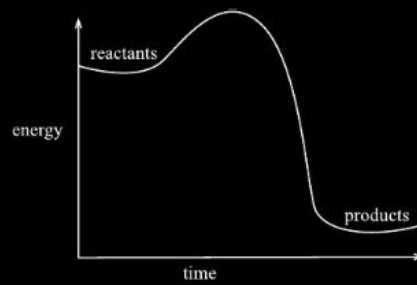


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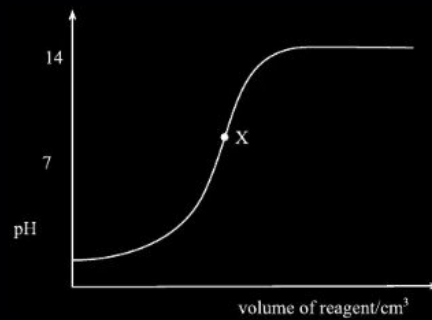
- 37 The graph shows how the energy changes as a reaction progresses.



Which statements about the reaction is/are correct?

1. the reaction is exothermic
2. the reaction is energetically feasible
3. the reaction is endothermic

- 38 The graph shows a titration curve.



Which statement(s) about the titration is/are correct?

1. A strong acid is titrated with a strong base.
2. X is the equivalent point
3. The base is being added to the acid.

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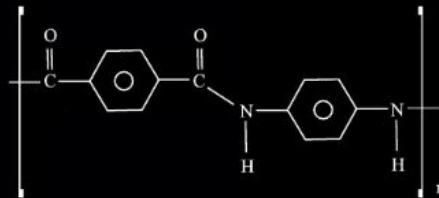


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39 The catalytic effect of nano particles is due to their

1. large surface area to volume ratio
2. nanometer-scale sizes
3. low melting points

40 The structure of Kevlar is shown.



Which statement(s) about Kevlar is/are correct?

1. it is a condensation polymer
2. consists of two different monomers
3. It is a polyamide

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